

EKF, UKF, AND PARTICLE FILTER COMPARISON FOR BALLISTIC REENTRY TRACKING: EFFECTS OF HEAT SHIELD ABLATION AND ATMOSPHERIC DISTURBANCES

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The Artemis II mission in April 2026 brought a renewed interest in the challenges of ballistic reentry vehicle tracking, where very nonlinear dynamics and extreme thermal conditions make accurate state estimation critical to the safety of the astronauts inside. This paper compares three nonlinear filtering approaches the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and the Particle Filter (PF), all applied to a simulated ballistic reentry problem. Two novel elements incorporated into this problem are a time varying ballistic coefficient driven by heat shield ablation, and a Gauss-Markov process modeled atmospheric wind. Throughout this experiment we showed that the UKF is the optimal filter for tracking the vehicle, providing a significantly lower cost to run than the PF, while being 10-50x more accurate than the simpler EKF.

INTRODUCTION

On April 10, 2026 millions across the US and the world were at the edge of their seats watching as NASA's Orion capsule carrying the four astronauts of the Artemis II mission plunged into the Earth's atmosphere at 35 times the speed of sound,⁴ enduring a six minute communication blackout as superheated plasma completely engulfed the vehicle. With no ability to change anything about reentry during those 6 minutes the math behind this process needs to be extremely well understood. As the vehicle's exterior fought back temperatures hotter than the sun's core, everyone was trusting that the math performed in the years leading up to this mission would hold up and keep the 4 astronauts safe.

This is the estimation problem at the heart of ballistic reentry tracking. The vehicle's dynamics are governed by a combination of nonlinear drag forces that depend on changing atmospheric density, varying exponentially with altitude, and a ballistic coefficient β that constantly changes as the vehicles heat shield erodes due to the temperatures experienced during reentry. On top of the non-linear dynamics, measurements from ground-based radar stations are themselves nonlinear functions of the vehicle's position. As such standard linear estimation techniques are wholly inadequate when it comes to problems like this.

This paper focuses on a comparative study of three nonlinear filtering approaches. These filters are the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and the Particle Filter (PF). We will be applying these three filters to a ballistic reentry tracking problem that not only models the nonlinear dynamics of the system, but also includes two new elements so that we may better capture the real world impacts of these different filters. These two elements are a time-varying ballistic coefficient, modeled by a change in weight of the vehicle due to ablation of its heat shield, as well as wind disturbances modeled as a Gauss-Markov process. Dual ground radar sensor fusion is used to improve state observability as well as β estimation accuracy.

In exploring this problem we hypothesize that the UKF and PF will outperform the EKF in tracking the ballistic reentry vehicle, as the strong nonlinearities in both the drag dynamics as well as the measurement model exceed the ability of a first-order linearization like the EKF.

The remainder of this paper is organized as follows: Section II provides the background and motivation, Section III details the methodology including our system dynamics and filter implementation, Section IV presents our simulation results, and Section V provides a discussion and conclusion.

BACKGROUND & MOTIVATION

The large nonlinearities of ballistic reentry dynamics present a fundamental challenge when performing state estimation. The governing equations of this system are highly nonlinear, and the nonlinearity only intensifies as the vehicle descends deeper into a denser atmosphere. The standard Kalman Filter (KF), while the optimal estimation method for linear systems, assumes that both the dynamics and the measurement model be strictly linear, which are both fundamentally violated in this problem.¹ The Linearized Kalman Filter (LKF) partially addresses this by linearizing the system about a precomputed nominal trajectory, but this approach quickly becomes inaccurate as the true trajectory deviates from the nominal, which in a system with growing nonlinearities, such as the one explored in this paper, is inevitable.

The first filter we explore in this paper is the EKF. This improves upon the LKF by linearizing the system at each time step, about the current estimate of the state. It accomplishes this by computing the Jacobian matrices, F ($\partial f / \partial x$) and H ($\partial h / \partial x$), at the current state estimate.² This filter is very computationally cheap making it a good option for mildly nonlinear systems, but linearization error quickly grows as a system becomes increasingly nonlinear, and for a system as complex as re-entry this filter is likely to be the least accurate of the three explored in this paper.

The second filter is the UKF, which completely avoids linearizing the problem by using a sigma points, a set of points that capture the mean and covariance of the current state estimate. These points are then propagated through our full nonlinear dynamics, which increases accuracy compared to LKF or EKF which treat the dynamics as though they are linear.² This filter properly captures our mean and covariance up to the third order for Gaussian distributions, compared to only first order accuracy for the EKF. As this filter doesn't require linearization, it is easier to implement for complex systems as we can avoid computing the Jacobian, and it better handles the growing nonlinearities of a system like reentry. The sigma points make the UKF slightly more computationally expensive than the EKF, but the improvements in accuracy for complex system, make this tradeoff worthwhile for highly nonlinear systems such as reentry.

The final filter that we will explore is the PF. This is the most complex of the filters, but also makes no assumptions on the linearity or shape of the error distribution, instead representing the full probability distribution as a cloud of random samples called particles.² Each particle is a guess of the what the current state might be, and they are each propagated individually through the nonlinear dynamics. This filter is more computationally intensive than the other two filters due to the propagation of particles which can range from 100s to 1000s depending on how complex the system is. This high cost is what makes it extremely accurate, which makes it likely to be the most accurate method for tracking the reentry vehicle that will be explored in this paper.

Prior work in ballistic reentry modeled the reentry vehicle assuming a constant ballistic coefficient and ignoring any effects from the wind,³ this paper improves upon prior work by incorporating

a changing ballistic coefficient, as well as a first order Gauss-Markov process for the wind. Additionally we incorporate measurements from 2 separate radar stations, improving state observability and reflecting the redundant sensor architectures used in real tracking systems.

As the vehicle descends, the atmospheric density increases exponentially, causing the drag forces to grow rapidly. Since drag scales with both density and the square of velocity, the nonlinearity of the system intensifies throughout the flight. The ever changing nonlinearity makes for the perfect benchmark for comparing the different filtering approaches, as the three filters are expected to show increasingly different performance as the vehicle approaches the ground.

METHODOLOGY

Problem Formulation & Assumptions

This paper focuses on the problem of tracking a ballistic reentry vehicle as it descends through the atmosphere, starting at an initial altitude of 90 km and ending at ground impact. Our primary objective is to estimate the vehicle's state, which in our case includes, position, velocity, and the ballistic coefficient, using noisy measurement data from two ground-based radar stations. Our simulation captures the aerodynamic effects on the vehicle during descent as well as the effects of heat shield ablation on the ballistic coefficient.

For the purposes of this paper, the following assumptions were made:

1. The vehicle was modeled as a point mass and rotational dynamics are neglected as they do not have a meaningful impact on the estimation problem, as the dominant nonlinearities arise from the aerodynamic effects on the vehicle and not its geometry.
2. Our vehicle's motion is restricted to two dimensions, those being horizontal and vertical. The addition of a third dimension introduces additional state variables without increasing the complexity of our estimation problem.
3. Atmospheric density varies exponentially with altitude according to

$$\rho(z) = \rho_0 e^{-z/H} \quad (1)$$

where $\rho_0 = 1.225 \text{ kg/m}^3$ is the air density at sea level and $H = 8500 \text{ m}$ is the atmospheric scale height.

4. The positions of both radar stations are exactly known.
5. The ablation rate of the heat shield is known for the truth simulation, but is withheld from the filter. This allows us to have realistic ablation in our truth simulation while allowing the filter to represent a realistic scenario where the exact degradation of the heat shield cannot be directly modeled, and must therefore be estimated.

Equations of Motion

In this paper our reentry vehicle is modeled as a point mass subject to two forces, gravity and aerodynamic drag. Gravity acts purely in the z direction

$$F_g = -mg\hat{z} \quad (2)$$

where $m = 1000$ kg (consistent with historical reentry vehicles), and g is the standard 9.81 m/s^2 . Next we have drag which opposes our velocity vector, $v = \sqrt{\dot{x}^2 + \dot{z}^2}$, with the unit vector opposing motion given by

$$\hat{u}_D = -\frac{\dot{x}\hat{x} + \dot{z}\hat{z}}{v} \quad (3)$$

and its magnitude is given by

$$F_D = \frac{\rho v^2 m}{2\beta} \quad (4)$$

where $\beta = m/(C_D A)$ is the ballistic coefficient in kg/m^2 , representing the ratio of vehicle mass to its aerodynamic drag area, and ρ is the atmospheric density given by Equation 1. Note that β is treated as an unknown quantity and augmented into the state vector for estimation, as detailed in Section . Applying Newton's second law in each direction and dividing by m , the equations of motions are:

$$\ddot{x} = -\frac{\rho v \dot{x}}{2\beta} \quad (5)$$

$$\ddot{z} = -\frac{\rho v \dot{z}}{2\beta} - g \quad (6)$$

The full nonlinear state equation $\dot{\mathbf{x}} = f(\mathbf{x})$ is:

$$\dot{\mathbf{x}} = f(\mathbf{x}) = \begin{bmatrix} \dot{x} \\ \dot{z} \\ -\frac{\rho v \dot{x}}{2\beta} \\ -\frac{\rho v \dot{z}}{2\beta} - g \\ -k_{abl} + w_\beta \end{bmatrix} \quad (7)$$

This system is highly nonlinear due to the multiplicative relation between states in the drag term, the exponential dependence of ρ on z , and the product of v and $\dot{x}$, $\dot{z}$. All these effects intensify as the vehicle descends deeper into the atmosphere, making this an increasingly nonlinear estimation problem throughout the flight.

Heat Shield Ablation Model

Ablation occurs due to the high temperatures experienced by the vehicle during reentry. As it travels through the atmosphere friction between the vehicle and the air superheats it, causing the decay of the heat shield slowly reducing its mass. On average a vehicle might lose $\sim 5\%$ of its mass during the entire reentry cycle. For a vehicle mass of 1000 kg this equates to a total loss of 50 kg, so for a travel time of approximately 120 seconds, this equates to a loss rate of 0.5 kg/s. Using $\beta = m/(C_D A)$ we can convert this to a value for our change in beta of $\dot{\beta} = 0.5$ $kg/m^2/s$. Real life ablation is never so perfect so to account for this we introduce some noise into our mass loss, we chose a noise value of $\sigma_\beta = 1.5$ $kg/m^2/s$, giving the full ablation model as:

$$\dot{\beta} = -k_{abl} + w_\beta, \quad w_\beta \sim \mathcal{N}(0, \sigma_\beta^2) \quad (8)$$

where $k_{abl} = 0.5$ $kg/m^2/s$ is the nominal ablation rate and $\sigma_\beta = 1.5$ $kg/m^2/s$ captures the stochastic uncertainty around that nominal rate. While the truth simulation propagates β according to Equation 8, the filter has no knowledge of this equation, and models the ballistic coefficient as a constant with $\dot{\beta} = 0$, with some process noise to allow it to slowly vary the value. This intentional model

mismatch simulates a realistic scenario where the exact heat shield degradation cannot be measured. This serves as a stress test of each filter's ability to track a slowly drifting parameter, without any knowledge of its governing equations.

Wind Disturbance Model

To differentiate this paper from prior work in state estimation of a ballistic reentry vehicle, we chose to add wind contributions to our simulated truth data. Instead of simply using randomized wind gusts that are completely independent, we chose to use the Gauss-Markov model for wind gusts. This ensures gusts are correlated in time instead of just randomly appearing and disappearing. The Gauss-Markov equations for wind gusts in the x and z directions are as follows:

$$\dot{w}_x = -\frac{w_x}{\tau} + n_x, \quad n_x \sim \mathcal{N}(0, \sigma_w^2) \quad (9)$$

$$\dot{w}_z = -\frac{w_z}{\tau} + n_z, \quad n_z \sim \mathcal{N}(0, \sigma_w^2) \quad (10)$$

where $\tau = 20$ s is the gust correlation time and $\sigma_w = 5$ m/s² is the gust intensity. We expect the effect of wind to be quite small on reentry relative to drag and gravity. Regardless of its magnitude we believe that including wind further creates a mismatch between our model and filter, further stressing each filter's ability to handle unmodeled disturbances. This model for wind is included in our truth simulation, but our filters remain completely unaware of the governing equations. The effect of wind is instead captured implicitly through the process noise covariance matrix $\mathbf{Q}$, which is tuned to reflect the expected magnitude of unmodeled disturbances.

Measurement Model

To measure the reentry vehicle we chose two ground based radars with known positions, one at the origin, (0,0) and one 300 km away in the x direction (300, 0). Two radars were chosen to better mimic the redundant measurement environments of real life systems, as for systems where small inaccuracies can lead to disaster, relying on a single source of measurements is just not very realistic. The dual radar configuration improves state observability, providing four measurements per timestep to estimate the five states in our system. Each radar measures the straight line distance to the vehicle (range) and the elevation angle above the horizon. The measurement equations for both radars are as follows:

$$r_1 = \sqrt{(x - x_{r1})^2 + z^2} \quad (11)$$

$$\theta_1 = \text{atan2}(z, |x - x_{r1}|) \quad (12)$$

$$r_2 = \sqrt{(x - x_{r2})^2 + z^2} \quad (13)$$

$$\theta_2 = \text{atan2}(z, |x - x_{r2}|) \quad (14)$$

where $x_{r1} = 0$ m and $x_{r2} = 300,000$ m are the known radar positions. The absolute value in the elevation angle equations ensures that θ remains between 0 and 90 regardless of which side of the radar the vehicle is on, consistent with how real radar systems report elevation angle. The full nonlinear measurement vector is therefore:

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} r_1 \\ \theta_1 \\ r_2 \\ \theta_2 \end{bmatrix} \quad (15)$$

Gaussian measurement noise is added to each measurement, with a standard deviation of $\sigma_r = 100$ m for range and $\sigma_\theta = 0.01$ rad ($\approx 0.57^\circ$) for elevation angle. These values were picked to be consistent with the performance of a typical ground based radar tracking system. With these values our measurement noise covariance matrix ends up being:

$$\mathbf{R} = \begin{bmatrix} \sigma_r^2 & 0 & 0 & 0 \\ 0 & \sigma_\theta^2 & 0 & 0 \\ 0 & 0 & \sigma_r^2 & 0 \\ 0 & 0 & 0 & \sigma_\theta^2 \end{bmatrix} \quad (16)$$

Note that all measurements are assumed to be independent, resulting in a diagonal $\mathbf{R}$ matrix.

Filter Implementation

Extended Kalman Filter The EKF relies on the linearization of the problem using the current best estimate of the state, using the following matrices:

$$\mathbf{F} = \frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & \frac{\rho_0 e^{-z/H} \dot{x} v}{2\beta H} & -\frac{\rho v}{2\beta} - \frac{\rho \dot{x}^2}{2\beta v} & -\frac{\rho \dot{x} \dot{z}}{2\beta v} & \frac{\rho v \dot{x}}{2\beta^2} \\ 0 & \frac{\rho_0 e^{-z/H} \dot{z} v}{2\beta H} & -\frac{\rho \dot{x} \dot{z}}{2\beta v} & -\frac{\rho v}{2\beta} - \frac{\rho \dot{z}^2}{2\beta v} & \frac{\rho v \dot{z}}{2\beta^2} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (17)$$

$$\mathbf{H} = \frac{\partial h}{\partial \mathbf{x}} = \begin{bmatrix} \frac{x-x_{r1}}{r_1} & \frac{z}{r_1} & 0 & 0 & 0 \\ \frac{z(x-x_{r1})}{r_1^2 |x-x_{r1}|} & \frac{-|x-x_{r1}|}{r_1^2} & 0 & 0 & 0 \\ \frac{x-x_{r2}}{r_2} & \frac{z}{r_2} & 0 & 0 & 0 \\ \frac{z(x-x_{r2})}{r_2^2 |x-x_{r2}|} & \frac{-|x-x_{r2}|}{r_2^2} & 0 & 0 & 0 \end{bmatrix} \quad (18)$$

where $\rho = \rho_0 e^{-z/H}$, $v = \sqrt{\dot{x}^2 + \dot{z}^2}$, and columns 3–5 of $\mathbf{H}$ are zero as measurements depend only on position, not velocity or β . The $\mathbf{F}$ Jacobian is the matrix of partial derivatives of the dynamics with respect to each state, and will be utilized in the propagation of the covariance, whereas the $\mathbf{H}$ Jacobian is the matrix of partial derivatives of the measurement function with respect to each state, and will be used in the EKF update step. The EKF consists of two main steps, the time update where the previous estimate is used to propagate the *a priori* state and covariance:

$$\hat{\mathbf{x}}_k^- = f(\hat{\mathbf{x}}_{k-1}) \quad (19)$$

$$\mathbf{P}_k^- = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^\top + \mathbf{Q} \quad (20)$$

where $\mathbf{Q}$ is the process noise covariance matrix, defined in Section ???. The second step is the measurement update, where the Kalman gain $\mathbf{K}_k$ is calculated and used to correct the *a priori* state using the innovation ($\mathbf{y}_k - h(\hat{\mathbf{x}}_k^-)$), which represents the difference between the actual and predicted measurement. The Kalman gain optimally weights the contribution of the innovation to the state update, balancing trust between the model prediction and the incoming measurement:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^\top (\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^\top + \mathbf{R})^{-1} \quad (21)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - h(\hat{\mathbf{x}}_k^-)) \quad (22)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \quad (23)$$

This process repeats at every time step so the filter can continue to refine its estimate as new radar measurements arrive. The EKF's reliance on linearization via the two Jacobian matrices makes it computationally efficient, but as previously discussed, this approximation is likely to degrade as the nonlinear reentry dynamics intensify as the vehicle approaches the ground.

Unscented Kalman Filter The benefit of the UKF over the EKF is that it entirely avoids linearizing our problem, by propagating a set of $2n$ sigma points through the full nonlinear dynamics, where $n = 5$, the dimension of our state vector. These sigma points are generate from our current best estimate of the state and covariance as follows:

$$\sigma_k^{(i)} = \hat{\mathbf{x}}_{k-1} + \left(\sqrt{n\mathbf{P}_{k-1}} \right)_i, \quad i = 1, \dots, n \quad (24)$$

$$\sigma_k^{(i+n)} = \hat{\mathbf{x}}_{k-1} - \left(\sqrt{n\mathbf{P}_{k-1}} \right)_i, \quad i = 1, \dots, n \quad (25)$$

where $\left(\sqrt{n\mathbf{P}_{k-1}} \right)_i$ denotes the i -th column of the Cholesky decomposition of $n\mathbf{P}_{k-1}$. Each of the sigma points is then propagated through the full nonlinear dynamics:

$$\sigma_k^{(i)-} = f \left(\sigma_{k-1}^{(i)} \right), \quad i = 1, \dots, 2n \quad (26)$$

The sigma points are the used to calualte the *a priori* state and covariance:

$$\hat{\mathbf{x}}_k^- = \frac{1}{2n} \sum_{i=1}^{2n} \sigma_k^{(i)-} \quad (27)$$

$$\mathbf{P}_k^- = \frac{1}{2n} \sum_{i=1}^{2n} \left(\sigma_k^{(i)-} - \hat{\mathbf{x}}_k^- \right) \left(\sigma_k^{(i)-} - \hat{\mathbf{x}}_k^- \right)^\top + \mathbf{Q} \quad (28)$$

The measurement update has a similar structure to the EKF, but instead of using the H Jacobian, each sigma point is put through the nonlinear measurement function directly:

$$\gamma_k^{(i)} = h \left(\sigma_k^{(i)-} \right), \quad i = 1, \dots, 2n \quad (29)$$

The predicted measurement, innovation covariance, and cross covariance are then computed as:

$$\hat{\mathbf{y}}_k = \frac{1}{2n} \sum_{i=1}^{2n} \gamma_k^{(i)} \quad (30)$$

$$\mathbf{P}_{yy} = \frac{1}{2n} \sum_{i=1}^{2n} \left(\gamma_k^{(i)} - \hat{\mathbf{y}}_k \right) \left(\gamma_k^{(i)} - \hat{\mathbf{y}}_k \right)^\top + \mathbf{R} \quad (31)$$

$$\mathbf{P}_{xy} = \frac{1}{2n} \sum_{i=1}^{2n} \left(\sigma_k^{(i)-} - \hat{\mathbf{x}}_k^- \right) \left(\gamma_k^{(i)} - \hat{\mathbf{y}}_k \right)^\top \quad (32)$$

The Kalman gain is then calculated without the need for any Jacobian, and finally we update the state and covariance in the same was as was done in the EKF:

$$\mathbf{K}_k = \mathbf{P}_{xy} \mathbf{P}_{yy}^{-1} \quad (33)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - \hat{\mathbf{y}}_k) \quad (34)$$

$$\mathbf{P}_k = \mathbf{P}_k^- - \mathbf{K}_k \mathbf{P}_{yy} \mathbf{K}_k^\top \quad (35)$$

This process repeats at every timestep, with the UKF continuously refining its estimate without ever needing to linearize the dynamics or measurement model.

Particle Filter The PF is a representation of the full posterior probability distribution of the states, which uses a set of N weighted particles, which are a random sample of our system. These particles make no assumptions about the linearity of the system or the shape of the error distribution, which is the PF's primary benefit over the previous two filters, which assume gaussian distributed noise. To start, N particles are created from an initial distribution centered around the initial state estimate:

$$\chi_0^{(i)} \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0), \quad i = 1, \dots, N \quad (36)$$

At each following timestep, each particle is individually propagated through the full nonlinear dynamics, with an added noise term:

$$\chi_k^{(i)-} = f\left(\chi_{k-1}^{(i)}, \mathbf{w}_{k-1}^{(i)}\right), \quad \mathbf{w}_{k-1}^{(i)} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (37)$$

As we get measurements, the likelihood q_i of each particle is computed by evaluating how well the particle's predicted measurement matches the actual radar measurement, as follows:

$$q_i = \frac{1}{(2\pi)^{m/2} |\mathbf{R}|^{1/2}} \exp\left(-\frac{1}{2} \left(\mathbf{y}_k - h\left(\chi_k^{(i)-}\right)\right)^\top \mathbf{R}^{-1} \left(\mathbf{y}_k - h\left(\chi_k^{(i)-}\right)\right)\right) \quad (38)$$

where m is the dimension of the measurement vector. These likelihoods are then normalized so they all add to one:

$$\tilde{q}_i = \frac{q_i}{\sum_{j=1}^N q_j} \quad (39)$$

We then perform a resampling by selecting N new particles from the current set according to their normalized likelihoods, keeping particles with higher likelihoods. For each new particle i , a random number, $r \sim \mathcal{U}[0, 1]$, is generated, and the smallest index j which satisfies $\sum_{m=1}^j \tilde{q}_m \geq r$ is found. This gives us:

$$\chi_k^{(i)+} = \chi_k^{(j)-} \quad (40)$$

The final state estimate is then calculated by finding the mean of all resampled particles:

$$\hat{\mathbf{x}}_k = \frac{1}{N} \sum_{i=1}^N \chi_k^{(i)+} \quad (41)$$

For this paper we use $N = 5000$ particles, which should provide a good balance between estimation accuracy and the computational cost for a 5 state system.

Simulation Setup

The true simulated data for this project is generated using a fixed-step fourth-order Runge-Kutta (RK4) integrator, with a time step of $dt = 0.1$ s. The fixed time step ensures that our wind and ablation noise is consistently injected into the system at each time step, this requirement makes a typical variable-step integrator such as ode45 not work. The simulation then runs from $t = 0$ until it reaches the ground, that being at $z = 0$.

Each filter is initialized with an imperfect estimate of the true initial state to help in simulating realistic operating conditions, where this initial position is not perfectly known. For our initial ballistic coefficient estimate we randomly select it from a distribution centered around the true value with a 20% uncertainty:

$$\hat{\beta}_0 = \beta_{true} (1 + 0.2 * \mathcal{N}(0, 1)) \quad (42)$$

The initial state covariance $\mathbf{P}_0$ was chose to reflect the uncertainty in our initial state estimate:

$$\mathbf{P}_0 = \text{diag} ([1000^2, 1000^2, 200^2, 200^2, 100^2]) \quad (43)$$

Our process noise matrix $\mathbf{Q}$ was designed to help account for the unmodeled wind disturbance as well as the ballistic coefficient drift that the filter was not made aware of:

$$\mathbf{Q} = \text{diag} ([0.1^2, 0.1^2, 5^2, 5^2, 2^2]) \quad (44)$$

Finally our measurement noise matrix $\mathbf{R}$ is set to be consistent with the noise levels used to generate the truth measurements:

$$\mathbf{R} = \text{diag} ([\sigma_r^2, \sigma_\theta^2, \sigma_r^2, \sigma_\theta^2]) \quad (45)$$

All of our simulation parameters are summarized in Table 1.

Table 1. Simulation Parameters

Parameter	Symbol	Value	Units
Gravitational acceleration	g	9.81	m/s ²
Sea-level air density	ρ_0	1.225	kg/m ³
Atmospheric scale height	H	8500	m
True ballistic coefficient	β_{true}	500	kg/m ²
Ablation rate	k_{abl}	0.5	kg/m ² /s
Ablation noise	σ_β	1.5	kg/m ² /s
Wind correlation time	τ	20	s
Wind gust intensity	σ_w	5	m/s ²
Radar 1 position	x_{r1}	0	m
Radar 2 position	x_{r2}	300,000	m
Range noise	σ_r	100	m
Angle noise	σ_θ	0.01	rad
Entry altitude	z_0	90,000	m
Entry speed	V	6000	m/s
Flight path angle	γ	-20	deg
Integration timestep	dt	0.1	s
Number of particles	N	5000	–

RESULTS

All three of the filters tested in this paper were evaluated on the same truth trajectory generated by the simulation previously described in this paper. Each filter was initialized with the same initial state estimate and covariance, with the ballistic coefficient initialized with a 20% error relative to the true value. To assess the performance of each filter, we used a Root Mean Square Error (RMSE) calculation between the filter estimate and the truth trajectory across all five states.

Extended Kalman Filter

Figure 1 shows the EKF results. From just looking at the trajectory in Figure 1(a) it appears as though the EKF does a very good job of tracking the true trajectory, however, the position and velocity error plots in Figures 1(c) and 1(d) help reveal that the estimation errors are actually a lot larger than they might seem in the trajectory plot. In those plots, we can see that our estimation error greatly increases as the vehicle descends deeper into the atmosphere. This is just what we expected as this problem's nonlinearities become greater the deeper in the atmosphere the vehicle is. The first order linearization of the EKF is capable of estimating the system while it is only slightly nonlinear as it first enters the atmosphere, but the ability of the filter to continue estimating rapidly degrades. Finally, Figure 1(b) shows the EKF's inability to estimate β as the value remains unchanged throughout the entire flight

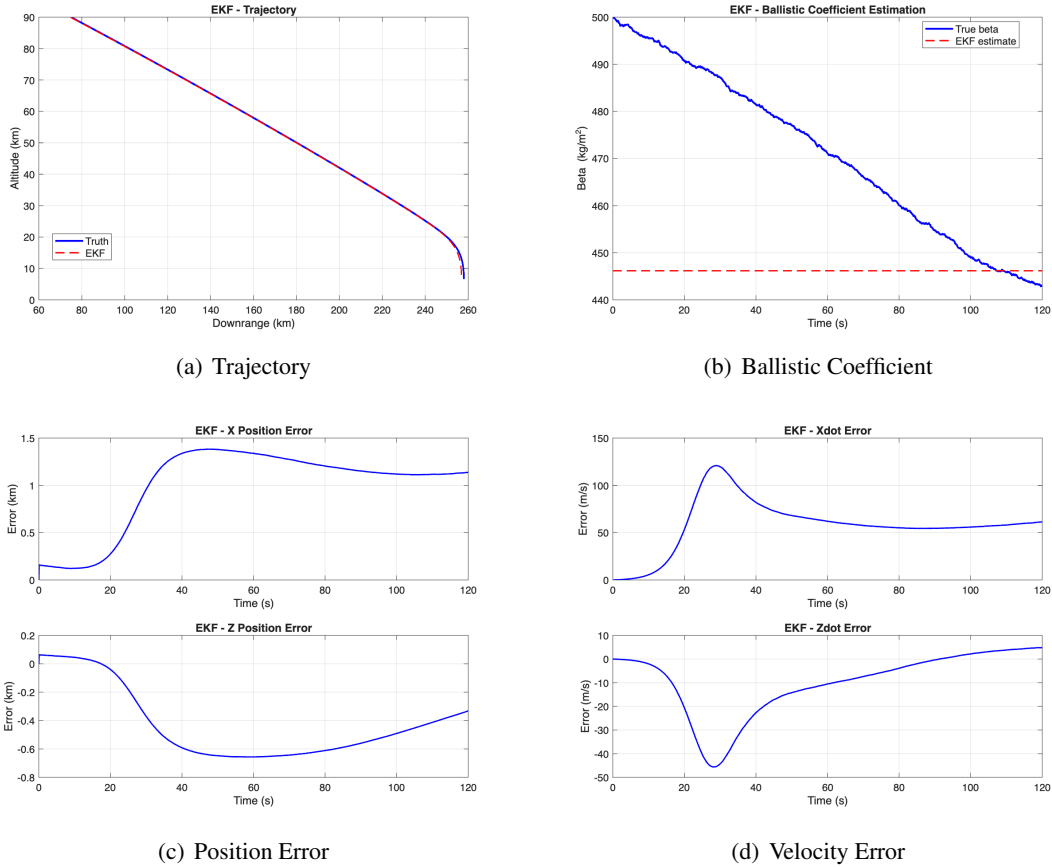


Figure 1. EKF results.

Unscented Kalman Filter

Figure 2 shows the UKF results. The trajectory in Figure 2(a) closely tracks the true trajectory throughout the entire flight, without any visible deviations, no matter how deep into the atmosphere the vehicle plunged. The position and velocity errors in Figures 2(c) and 2(d) show very small and bounded errors throughout the entire flight, oscillating around a minimum error value instead of growing over time. Figure 2(b) shows that the UKF more quickly converges closer to the true

value of β , but after this initial convergence it struggles to track the continuing decrease, instead oscillating around a fixed value closer to what β was originally set to.

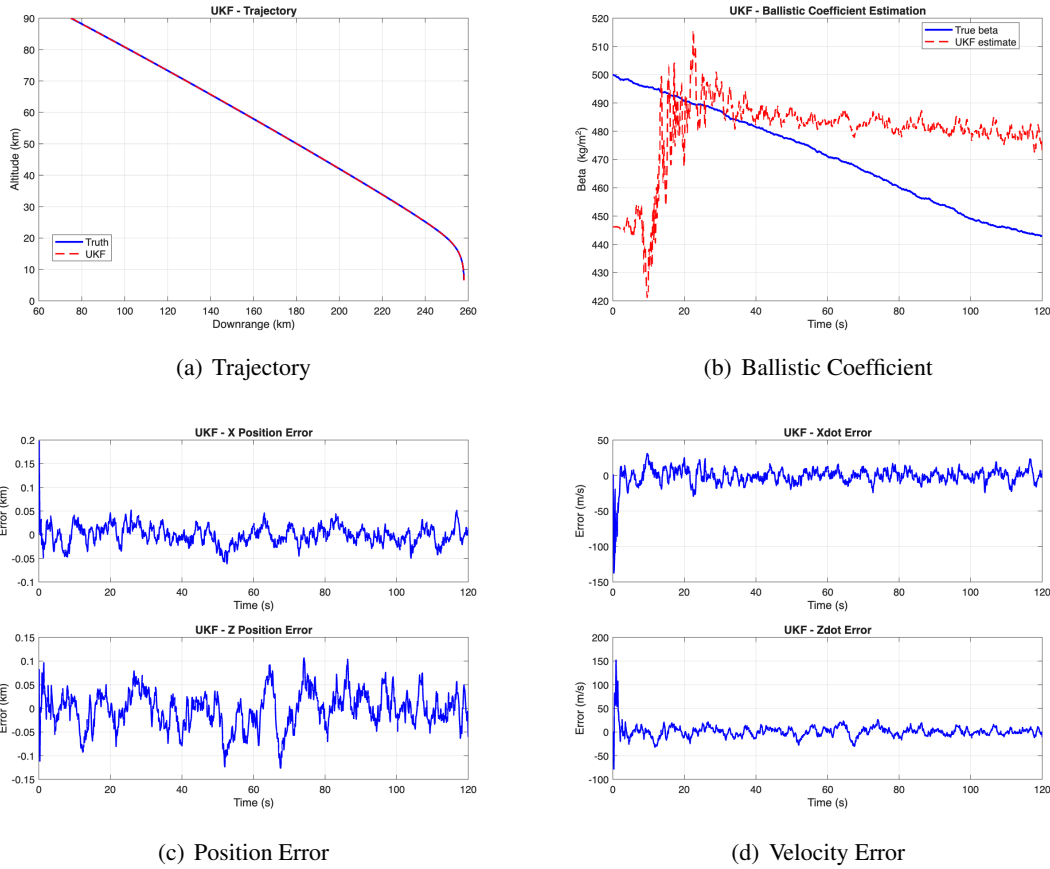
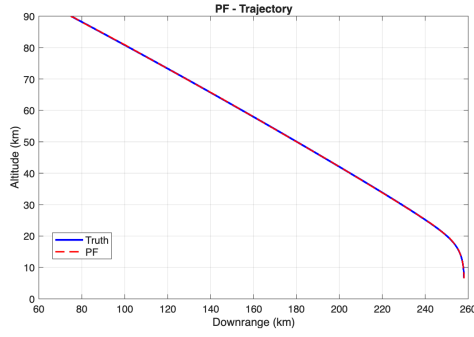


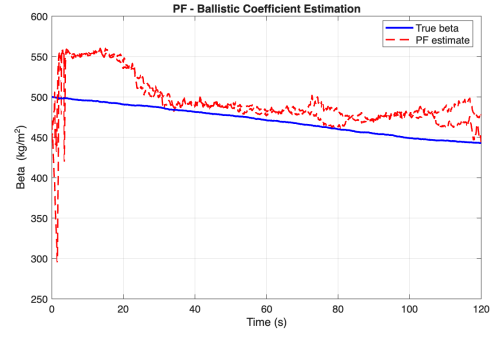
Figure 2. UKF results.

Particle Filter

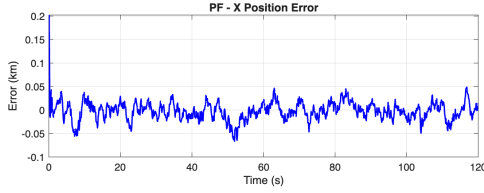
Figure 3 shows the PF results with $N = 5000$ particles. The trajectory in Figure 3(a), similar to the UKF, closely tracks the true trajectory throughout the entire flight without any noticeable deviations. The position and velocity error plots in Figures 3(c) and 3(d) also show comparable performance to the UKF, where the error slightly oscillates but is bounded and never has any large spikes in value. Figure 3(b) shows the PF estimation of the ballistic coefficient, which similarly to the UKF undergoes large initial oscillations showing the difficulty of estimating β in the first few seconds of flight, but it quickly converges close to the true value. It then is able to track the change in β over time, but is never able to perfectly estimate it.



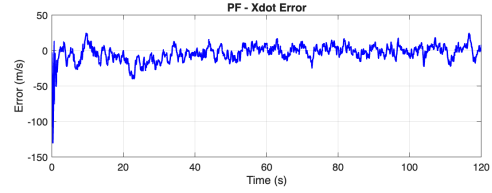
(a) Trajectory



(b) Ballistic Coefficient



(c) Position Error



(d) Velocity Error

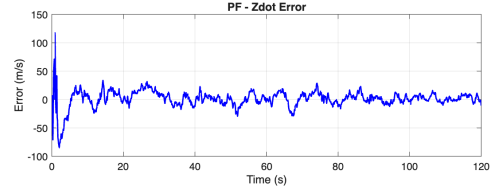
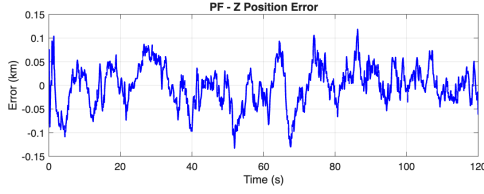


Figure 3. PF results.

DISCUSSION

Filter Performance Comparison

Table 2 summarizes the RMSE for all five states for the three filters.

Table 2. RMSE Comparison Across All Filters

Filter	x (m)	z (m)	$\dot{x}$ (m/s)	$\dot{z}$ (m/s)	β
EKF	1080.9	492.6	64.6	16.6	30.3
UKF	20.0	38.6	12.7	12.8	26.3
PF	19.6	42.1	12.1	14.7	31.8

EKF Performance

As expected, the EKF performs significantly worse than both the UKF and PF, across all states. The downrange position error of the EKF exceeds 1000 m, making it approximately 50x worse than the two other filters, with its performance across other states also being 2-10x worse. As we saw before the EKF's degradation is not uniformly distributed throughout the whole flight, errors remain

fairly low as the vehicle is in the upper atmosphere, where drag is negligible, but they rapidly grow as the vehicle reaches denser parts of the atmosphere where the nonlinearities rapidly increase. This confirms the hypothesis that the first-order linearization performed by the EKF are not enough to adequately account for the growing nonlinearities of a ballistic reentry vehicle. The EKF's lacking estimation of position and estimate also feed into it's estimate of β through the Kalman gain. This might be able to explain why the EKF is unable to actually estimate β , instead keeping it constant throughout the entire flight.

UKF vs. Particle Filter

The UKF and PF both greatly outperform the EKF, but relative to each other have very similar performance across all states. The UKF achieves slightly better altitude and β estimation, while the PF has slightly better downrange and $\dot{x}$ estimates. However, the differences between the two are extremely small and cannot be trivially attributed to differences between the filters, instead are most likely caused by the fact this was all from one single run of each filter. Both filters successfully avoid the divergence seen in the EKF through their propagation of the nonlinear dynamics without the need for linearization. As this problem has near-gaussian noise, it is understandable that both of these filters would have similar performances, as the main advantage of the PF is it's ability to deal with more complicated forms of noise, not present in this problem. Even though their performance was similar, the PF required $N = 5000$ particles leading to significantly more computational cost and run time than the UKF, bringing into question its usefulness for a problem like ballistic reentry.

Ballistic Coefficient Estimation

All three filters displayed significantly different behaviors in their estimates of β . The EKF remained completely static throughout the entire flight, never tracking the ablation driven decrease in β . The UKF and PF undergo very large oscillations during the initial 20-25 seconds of flight, as the filters attempt to identify the true value of β from our noisy radar measurements. After this initial period of high oscillations both filters converge to a value closer to what the true value of β is. The PF then is able to slightly track the decline in the value of β whereas the UKF remains essentially constant with small oscillations occurring around a fixed value. This behavior can be attributed to the process noise on β given in $\mathbf{Q}$ having been tuned conservatively preventing either filter from making massive changes to β after it initially got close to the estimate. Increasing σ_β might lead to better and faster tracking of β at the cost of increased estimate noise. Despite the fact that the EKF is unable to estimate β it still reaches a relatively low RMSE of 30.3, even lower than the PF, as, coincidentally, its initial value of β happens to land close to the final value of the true β . Out of all three the UKF achieves the most accurate RMSE of 26.3, due to its more accurate convergence early in flight.

Hypothesis Evaluation

The results confirm our hypothesis that the UKF and PF would significantly outperform the EKF due to the strong and ever growing nonlinearities present in the reentry drag dynamics. The EKF's failure is more severe than anticipated, as not only do its position and velocity estimates diverge, but its β estimate completely freezes, demonstrating that linearization errors propagate through the entire filter and can corrupt all state estimates. We initially hypothesizes that the PF would be the best filter to use for a problem like this but given the near equal if not better performance of the UKF, as well as the significantly reduced computational cost, as the PF required $N = 5000$ particles, these

experiments showed that the UKF might actually be the best option for tracking a ballistic reentry problem. This helped show that the most complex filter is not always the best option, as a simpler option might work just as well but also cost significantly less time and resources to run.

CONCLUSION

This paper presents a comparison of three nonlinear filtering approaches, the Extended Kalman Filter, the Unscented Kalman Filter, and the Particle Filter, each applied to a ballistic reentry vehicle tracking problem. In this paper our simulation of reentry incorporated two new problems: a time varying ballistic coefficient driven by heat shield ablation, and atmospheric wind disturbances modeled by a Gauss-Markov process. Two ground based radar sensors were fused to improve state observability as well as β estimation.

The results clearly demonstrate that the EKF is completely inadequate for this problem. The EKF's first order linearization quickly becomes unable to handle the nonlinearities of this problem, which rapidly grow as the vehicle reaches denser regions of the atmosphere where drag becomes too great to neglect. This was shown through the large position and velocity errors, but also through the filter's inability to estimate β . The UKF and PF both avoid this problem by propagating the full nonlinear dynamics, letting it achieve position errors of about 20 m compared to the EKF's over 1000 m.

The UKF and PF performed comparably across all the states, confirming the ballistic reentry's near gaussian noise distribution can be represented by both the UKF's sigma points as well as the PF particle cloud. The PF's ability to represent more complex distributions of noise present no benefit here, and as it required 5000 particles to match the performance of the UKF, which only used 10 sigma points, the PF large computational cost makes it the worst of the two for estimating this problem. All three filters struggle to track the continuing decline of β , but future tuning of $\mathbf{Q}$ could result in an increase in the filter's ability to track β .

Overall, this experiment showed that the UKF is the optimal choice for tracking the ballistic reentry vehicle. It achieves the accuracy of the PF at a significantly reduced computational cost, and vastly outperforms the EKF under the growing nonlinearities of atmospheric reentry.

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